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Co-Processing of Used Cooking Oil in Hydrocracker: Refinery-Scale Trial for Sustainable Aviation Fuel Production

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Abstract

This study presents a field-scale demonstration of Used Cooking Oil (UCO) co-processing in a commercial hydrocracker, with the primary objective of evaluating its viability as a pathway for producing Sustainable Aviation Fuel (SAF) and renewable diesel. Key performance indicators assessed include reactor thermal behaviour, catalyst stability, biogenic carbon integration, and product quality. The trial supports the broader refinery objective of aligning with decarbonisation mandates using existing infrastructure and minimal capital investment.

Among the process units in a typical refinery, Hydrocrackers stand out for their versatility and superior adaptability to handle renewable bio-feeds. A controlled co-processing trial was executed in Hydrocracker Unit with UCO fed at up to 3 vol% of total feedstock. UCO, a second-generation biofeeds rich in long-chain triglycerides and oxygenates, was introduced via a filtered, dedicated pump system connected to the main feed line. Reactor bed temperatures, pressure drops, hydrogen partial pressure, off-gas composition, and sour water generation were monitored in real-time. Analytical evaluation of product fractions included cold flow properties, distillation behaviour, and biogenic carbon quantification using ASTM D6866. Particular emphasis was placed on temperature profiles across the catalyst beds, detection of trace gases, and water production behaviour to assess the extent of hydrogenolysis, decarboxylation, and hydrodeoxygenation reactions.

While hydrotreaters have been widely used for co-processing, hydrocracking units offer distinct advantages. Their multi-bed configuration, higher hydrogen partial pressures, and robust temperature management systems make them better suited to handle the operational challenges posed by biofeeds—notably high hydrogen demand, exothermic reactions, and the presence of contaminants. In addition, hydrocrackers have integrated fractionation systems, enabling effective separation of renewable-derived molecules into desired fuel product streams such as jet and diesel. The trial validated the hydrocracker's capability to process oxygenated renewable feedstocks under commercial operating conditions. A moderate exothermic rise was observed in reactor Bed-1 and Bed-2 due to hydrodeoxygenation, decarboxylation and saturation reactions, while downstream beds exhibited minimal impact, indicating rapid reaction kinetics.

of UCO components compared to fossil VGO. Gross conversion improved and middle distillate yield increased, with a product shift toward kerosene and diesel.

Biogenic carbon was successfully traced (ASTM D6866) in ATF fraction and diesel fraction, supporting the effectiveness of co-processing for renewable content integration. Though ATF freezing point and diesel pour point increased, still remained within product specification, with slight improvement in cetane index. These results affirm the viability of SAF production from co-processing routes using existing refinery infrastructure.

Introduction

The global aviation industry faces increasing pressure to decarbonize, with Sustainable Aviation Fuel (SAF) emerging as a key enabler in reducing lifecycle greenhouse gas (GHG) emissions. In alignment with international frameworks such as the Carbon Offsetting and Reduction Scheme for International Aviation (CORSIA), India has signalled its intent to adopt SAF through indicative blending targets set by the Ministry of Petroleum and Natural Gas—1% by 2027, 2% by 2028, and 5% by 2030 for international flights. However, large-scale SAF production remains a significant challenge due to limited infrastructure, high production costs, and the need for technological adaptability within existing refinery setups. Co-processing bio-based feedstocks, such as Used Cooking Oil (UCO), in conventional refining units like hydrotreaters and hydrocrackers, offers a cost-effective and scalable pathway to produce SAF with minimal modifications to existing assets.

This paper presents a refinery-scale feasibility study on the co-processing of UCO in the hydrocracker unit (HCU) at Mangalore Refinery and Petrochemicals Limited (MRPL). The HCU was evaluated as a candidate for SAF production using existing catalyst systems. A pilot plant study conducted by MRPL Innovation Centre confirmed that up to 5% UCO can be co-processed with straight run vacuum gas oil (SRVGO), enabling the production of a SAF blend compliant with ASTM D1655 specifications. Building on these results, a full-scale trial was undertaken to validate process operability, catalyst integrity, and product quality. The paper highlights feed characterization, catalyst performance, and SAF yield, offering a replicable model for other refineries aiming to integrate SAF production within their existing operations.

Feedstock Integration Challenges in Hydrocracker Co-Processing

As the global energy sector undergoes a transition toward lower-carbon solutions, the integration of renewable feedstocks into conventional refinery systems has become a strategic priority. While early efforts focused on first-generation feedstocks such as palm oil and other triglyceride-based oils, the industry is now exploring the incorporation of more complex and less refined second- and third-generation materials. These include waste-derived oils and non-edible bio-resources, which present distinct processing challenges due to their variable composition, higher impurity levels, and increased reactivity under hydroprocessing conditions.

Although co-processing renewable materials in hydrotreater units is now an accepted and proven pathway, the extension of this approach to hydrocrackers brings both opportunities and limitations. Hydrocrackers offer deeper conversion and can yield higher-value products such as kerosene-range fuels; however, their sensitivity to feedstock quality and the performance demands on catalysts make them less forgiving than hydrotreaters. Key technical concerns include potential impacts on catalyst life, fouling tendencies, increased hydrogen consumption, and possible shifts in product selectivity. These factors become even more significant when transitioning from relatively clean first-generation bio-oils to more challenging feedstocks like used cooking oil, algal oils, or pyrolysis oils, which may carry metals, oxygenates, or high levels of contaminants. Therefore, a thorough understanding of how these feedstocks interact with hydrocracker catalysts and affect overall unit performance is critical to assessing feasibility and developing robust co-processing strategies

Feedstock Integration Challenges & Process Chemistry

The transformation of renewable feedstocks into transportation fuels within conventional refinery units has gained momentum in recent years. Among the most widely processed renewable materials are triglyceride-based oils, primarily derived from first-generation feedstocks such as virgin vegetable oils. These compounds are structurally composed of three long-chain fatty acids attached to a glycerol backbone, forming ester linkages.

In recent developments, attention has shifted toward second-generation feedstocks—notably Used Cooking Oil (UCO) and Palm Oil Mill Effluent (POME)—which are by-products or waste streams from food or oleochemical industries. Third-generation biofeeds, such as pyrolysis oils derived from municipal solid waste, agricultural residues, and waste plastics or tyres, are also being explored for future integration, though they pose greater processing challenges due to their diverse chemical makeup.

During co-processing, the renewable feed is introduced alongside fossil-derived feedstocks (e.g., VGO) into hydroprocessing units such as hydrotreaters or hydrocrackers. In the case of triglyceride-rich biofeeds like UCO, the conversion chemistry involves several key steps:

- Saturation of unsaturated bonds within the fatty acid chains.
- Hydrodeoxygenation (HDO), where oxygen is removed as water (H₂O), carbon monoxide (CO), or carbon dioxide (CO₂) via reaction with hydrogen.
- C-C bond cleavage, resulting in the formation of propane from the glycerol backbone and long-chain n-paraffins from the fatty acid chains.

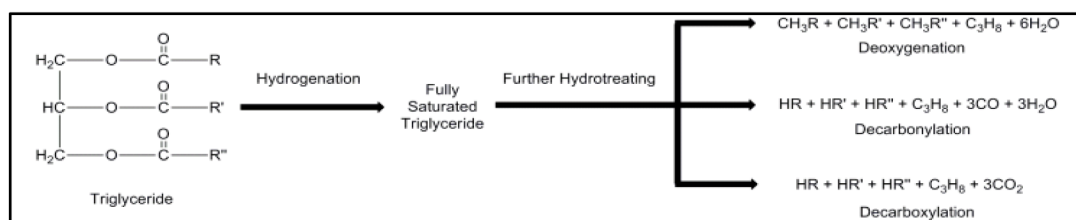
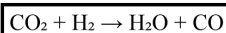


Figure 1—Process Chemistry of Triglyceride conversion in Hydrocrackers

Secondary reactions such as methanation and the reverse water-gas shift (RWGS) reaction are also expected to occur, leading to the formation of methane (CH₄) and carbon monoxide (CO) as byproducts during the hydroprocessing of bio-derived feedstocks

Reverse Water-Gas Shift (RWGS) Reaction



Methanation Reaction

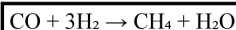


Figure 2—Process Chemistry of secondary reactions

These reactions are highly exothermic and hydrogen-intensive. The resulting products are generally linear paraffins that fall within the middle distillate boiling range, making them suitable for ATF & diesel applications.

Jet fuel-grade hydrocarbons require either chain shortening (hydrocracking) or molecular rearrangement (isomerisation) to adjust the cold flow properties and reduce the cloud point. Linear paraffins, although beneficial for high cetane diesel, exhibit poor cold flow behavior, which limits their direct use in aviation

fuels. Isomerisation, often facilitated by hydrocracking catalysts, modifies the straight chains into branched structures, improving cold flow characteristics without compromising on energy density.

When second- or third-generation biofeeds are involved, the chemistry becomes more complex. These materials may contain aromatics, heteroatoms, and oxygenated ring compounds, which are more resistant to hydrotreatment and tend to produce heavier naphthenic compounds. Effective processing of these requires selective ring opening and hydrocracking reactions to shift the product distribution toward the jet and diesel range and to meet fuel property specifications.

In the context of MRPL's hydrocracker co-processing trial, understanding the reactivity profile of UCO, its hydrogen demand, interaction with existing catalysts, and the exothermicity of reactions is crucial for maintaining operational stability and achieving the desired product quality—particularly SAF that complies with ASTM D1655 standards.

Operational Insights from UCO Co-Processing.

Pre-treated UCO. Pre-treated Used Cooking Oil was co-processed in one of our hydrocracker units at a concentration of 3 vol% of the total fresh feed. Pre-treatment of UCO is a critical prerequisite to mitigate risks associated with metallic contaminants such as silicon, iron, sodium, potassium, magnesium, and calcium, which can lead to catalyst poisoning and reduced reactor performance. Additionally, the presence of free fatty acids and chloride compounds pose serious corrosion risks—with FFAs affecting metallurgy in the feed preheat section and chlorides impacting the effluent handling system. Typical pre-treatment processes are

- Filtration : Removal of food particles & suspended solids
- De-gumming : Removal of phospholipids
- Dehydration : Removal of moisture
- Neutralization : Removal of free fatty acids
- Metals removal

To ensure safe and reliable operation, the pre-treated UCO was required to meet stringent specifications:

- Total metal contaminants limited to ≤ 10 wtppm
- Phosphorus content < 6 ppm
- Total Acid Number (TAN) < 0.5 mg KOH/g
- Free Fatty Acids < 1 wt%

These specifications were developed to ensure that, upon blending with the conventional fossil feed, the resulting combined feed remained within acceptable limits for both catalyst compatibility and equipment metallurgy, thereby enabling a successful coprocessing trial.

Process Condition & Catalyst Performance

Effect on Catalyst. During the UCO co-processing trial, lower catalytic severity was observed across both the pretreat and cracking beds of the hydrocracker, as indicated by a decrease in Weighted Average Bed Temperatures

- Pretreat Bed WABT dropped by 0.9°C , consistent with the expectation that hydrotreating reactions of triglycerides are less demanding than those required for complex fossil VGO molecules (e.g., hydrodesulfurization, hydrodenitrogenation, and hydrogenation of aromatics).

- The Cracking Bed Weighted Average Bed Temperature (WABT) decreased by 0.6°C, which is attributed to the presence of C14-C18 paraffinic hydrocarbons derived from UCO. These molecules predominantly fall within the distillate boiling range and undergo minimal further cracking, resulting in limited true conversion over the cracking catalyst.
- A distinct shift in exothermic behavior was observed: Exotherm increased across pretreat beds (by ~2–2.5°C), due to hydrogenation and hydrodeoxygenation of triglycerides. Exotherm decreased across cracking beds (up to –1°C), indicating lower cracking severity during co-processing.
- No significant change in reactor differential pressure was observed during the trial period, indicating stable flow dynamics and no impact on catalyst bed fouling or pressure drop under co-processing conditions

These trends confirm that triglyceride conversion occurs primarily in the pretreat section, and the reduced cracking demand aligns with the middle distillate range paraffinic nature of the UCO feedstock.

Effect on Conversion. The triglycerides present in UCO easily get converted entirely into products over catalyst in hydrocracker operating conditions. As a result, UCO does not contribute to unconverted oil fraction. Therefore the overall conversion of hydrocracker increased by approx. 1%.

Effect on Recycle gas Purity & Gas make. A decline in recycle gas purity accompanied by an increase in off-gas flow rates was observed during the co-processing trial. This behavior is consistent with the expected reaction pathways during triglyceride conversion. Specifically, C₃ hydrocarbons are generated through cleavage of the triglyceride backbone, methane (C₁) is formed via methanation reactions, and carbon dioxide (CO₂) results from decarboxylation processes. These reaction products were reflected in the elevated concentrations of C₁, C₃, and CO₂ observed in the sponge and amine absorber off-gases during the trial. Limited impact on recycle gas purity was observed during the short-duration trial. However, the effects on recycle gas composition during prolonged or continuous co-processing operations are yet to be seen and may warrant further evaluation

Effect on Chemical Hydrogen Consumption. Used Cooking Oil exhibits a higher hydrogen demand compared to fossil VGO, primarily due to the extensive hydrogenation required to convert its oxygenated and unsaturated components into stable hydrocarbons. When comparing the chemical hydrogen consumption between the baseline 100% fossil VGO operation and the 3% UCO co-processing run, a marginal increase of approximately 3.5 Nm³/m³ of feed was recorded.

Impact on downstream Units.

- Co-processing bio-feedstocks in conventional refineries leads to notable changes in downstream treating units. Increased oxygen content in the feed generates more CO₂ and water, shifting amine system feed gas composition toward relatively higher CO₂, which can increase regenerator energy demand, corrosion risk, and foaming tendency. When a selective amine such as MDEA is used, part of the additional CO₂ can bypass absorption in the amine absorber, resulting in less impact on the acid gas composition compared to non-selective amines. In certain cases, refineries may find it advantageous to replace MEA, DEA with MDEA to better manage higher CO₂ levels while also lowering energy requirements
- Sour water volume typically rises, requiring additional stripping capacity and energy, while higher organic acid and phosphate levels may introduce operational challenges.
- In the sulphur recovery unit, elevated CO₂ reduces reaction furnace temperature, potentially affecting contaminant destruction and requiring mitigation such as oxygen enrichment or co-firing. Tail gas treating units also see increased CO₂ loadings, necessitating adjustments in amine circulation or strength to maintain performance.

Product Yield. Used Cooking Oil (UCO), like most vegetable oils, is primarily composed of triglycerides derived from fatty acids. When hydrotreated during co-processing, these triglycerides are broken down into paraffinic hydrocarbons. The predominant carbon chain lengths in the paraffins formed from UCO depend on the original fatty acid composition, which varies slightly depending on the source of the oil (e.g., soybean, palm, canola). However, based on literature, generally UCO tends to produce paraffins in the C15 to C18 range, with C16 (hexadecane) and C18 (octadecane) being the most prominent. These correspond to the most common saturated fatty acids found in vegetable oils, namely:

- Palmitic acid (C16:0)
- Stearic acid (C18:0)

Upon hydrodeoxygenation, these fatty acids are converted into their respective n-paraffins, which are expected to predominantly fall within the kerosene and diesel boiling range based on their boiling points. The presence of paraffins influences both product yield and properties.

As expected, there was an increase in middle distillate yield (HSD+ATF) during co processing of Used Cooking Oil indicating that C16, C18 molecules from triglyceride side chain were retained in middle distillate. Overall C3 content in products increased marginally, primarily attributed to the breakage of backbone from the triglyceride.

Product Properties

- The UCO feed primarily contains triglycerides with C16 and C18 side chains as mentioned above. During hydro processing, these are fully saturated to form n-paraffins, which typically boil within the middle distillate range. However, n-paraffins are known for their poor cold flow properties, which can affect the cold flow performance of diesel and kerosene produced during co-processing. The boiling ranges of the resulting paraffins are provided below for reference.
- **Diesel** cold flow properties such as CFPP and pour point, typically show a decline, owing to the higher paraffin content. However, this increase in paraffinic content also contributes to an improvement in the cetane index, enhancing combustion quality.

During the trial, the expected deterioration in cold flow performance was observed, accompanied by a measurable increase in cetane index. These shifts in diesel quality are consistent with the nature of renewable feed conversion.

- **Kerosene:** As expected, the kerosene freeze point increased significantly during the co-processing run compared to the 100% fossil VGO feed case. A comparison of kerosene properties at 2.5% co-processing and the 100% fossil VGO case is provided below. These results are consistent with expectations, as co-processing increases the paraffin content in kerosene, leading to an increase in freeze point and reduction in aromatics content. The decrease in aromatics is further supported by the observed improvement in smoke point, indicating enhanced product quality.
- **Biogenic carbon analysis:**

Biogenic carbon comes from recent biological sources. It contains measurable amounts of carbon-14 (^{14}C), a radioactive isotope naturally present in living organisms. Fossil fuels (e.g., crude oil, coal) contain no detectable ^{14}C , because it decays completely over millions of years.

Table 1—Boiling points of normal Paraffins

Chain Length	Boiling Point, °C
nC14	254
nC15	271
nC16	287
nC17	303
nC18	317
nC19	330

Table 2—Variation in Diesel product properties

Quality parameter	Fossil Run	Co Processing Run
CFPP	Base	5°C 
Cetane Index	Base	1.2 

Table 3—Variation in Kerosene product properties

Quality parameter	Fossil Run	Co Processing Run
Freeze point	Base	15°C 
Smoke Point	Base	0.5 mm 

Radiocarbon Analysis (ASTM D6866)

ASTM D6866 - Standard Test Methods for Determining the Bio-based Content of Solid, Liquid, and Gaseous Samples Using Radiocarbon Analysis. EN 16640 - Equivalent European standard.

- Measure the proportion of carbon-14 (^{14}C) to total carbon in the sample.
- The ^{14}C content reflects the amount of bio-based carbon in a fuel product.

Based on ASTM D6866 radiocarbon analysis, the presence of biogenic carbon in the co-processed product streams was conclusively established. This standardized method enabled accurate quantification of renewable carbon content in both the kerosene (ATF) and diesel fractions, thereby confirming the incorporation of bio-derived carbon into the final products during the hydrocracker co-processing trial. The results indicated that the ATF stream contained approximately 2% biogenic carbon, and the diesel stream contained around 6%, effectively validating the in-situ production of Sustainable Aviation Fuel (SAF) and renewable diesel within the existing refinery infrastructure

Conclusions & Recommendations

The co-processing trial of Used Cooking Oil (UCO) in the Hydrocracker Unit (HCU) demonstrated that the unit is capable of handling renewable feedstocks with manageable operational impacts. Key observations, as compared to 100% fossil VGO operation, included:

- Reduced catalytic severity in both treating and cracking beds
- Increased overall conversion at lower severity
- Elevated off-gas generation and slightly lower recycle gas purity
- Marginally higher hydrogen consumption
- No significant impact on reactor differential pressure
- Clear evidence of hydrodeoxygenation (HDO) and decarbonylation (DCO) reactions through increased sour water generation and CO₂ presence
- A favorable shift in product yields toward middle distillates

The successful execution of the co-processing trial was made possible through the essential pre-treatment of Used Cooking Oil (UCO), which ensured safe operation, catalyst protection, and equipment integrity. Pre-treatment steps such as filtration, degumming, dehydration, neutralization, and metals removal were necessary to eliminate contaminants like metals, free fatty acids, phosphorus, and chlorides. These steps brought the bio-feed within the required specification limits—namely, ≤ 10 wtppm total metals, < 6 ppm phosphorus, < 0.5 mg KOH/g TAN, and < 1 wt% free fatty acids—making it suitable for blending with conventional VGO. This highlights that proper feedstock pre-treatment is essential for the reliable co-processing of renewable feedstocks in existing refinery units, enabling the production of SAF and renewable diesel without operational issues.

Co-processing bio-feedstocks in conventional refineries increases CO₂ and water production, impacting amine systems, sour water strippers, sulphur recovery units, and tail gas treating units. Higher CO₂ can raise energy demand, corrosion risk, and foaming, though selective amines like MDEA can mitigate some effects. Sour water volumes typically increase, necessitating extra stripping capacity, while organic acids and phosphates may create operational challenges. In SRUs, elevated CO₂ can lower furnace temperatures, requiring measures such as oxygen enrichment or co-firing, and TGTUs may need adjustments in amine circulation or strength to sustain performance.

The existing middle distillate-selective catalyst system successfully supported the co-processing of 2.5-3 vol% biogenic oil, which resulted in the production of approximately 2% biogenic carbon in ATF and 6% in diesel, while meeting all relevant SAF product quality specifications. These findings confirm the technical feasibility of producing Sustainable Aviation Fuel through co-processing in the current hydrocracker configuration, without requiring major modifications to equipment or catalyst system.

To increase SAF output beyond the current limit, catalyst reformulation—particularly the use of a dewaxing catalyst—would be necessary to meet cold flow property specifications like freeze point. In conclusion, this trial confirms that co-processing of renewable feedstocks such as UCO in hydrocrackers is a technically viable pathway for the production of Sustainable Aviation Fuel (SAF), enabling effective decarbonization of refinery operations within existing infrastructure while supporting India's alignment with global aviation emission reduction targets

Nomenclature

ASTM	- American Society for Testing and Materials
CORSIA	- Carbon Offsetting and Reduction Scheme for International Aviation
GHG	- Green House Gas
SAF	- Sustainable Aviation Fuel
HSD	- High Speed Diesel
VGO	- Vacuum Gas Oil
UCO	- Used Cooking Oil

WABT - Weighted Average Bed Temperature
HCU - Hydrocracker Unit
POME - Palm Oil Mill Effluent
HDO - Hydrodeoxygenation
FFA - Free Fatty Acids
Nm³ - Normal cubic meter
m³ - Cubic meter
°C - Degree Celsius
MDEA - Methyl Diethanolamine
MEA - Monoethanolamine
DEA - Diethanolamine
TAN - Total Acid Number
SRU - Sulphur Recovery Unit
TGTU - Tail Gas Treatment Unit
Wtppm - Weight in parts per million
Wt% - Weight percentage
g - Gram
mg KOH - milligram of Potassium hydroxide

Reference

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